Procura AI: A Stateful Multi-Agent Framework for Evidence-Aware Procurement Automation

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Abstract— Procurement spans requirement intake, supplier sourcing, quotation comparison, negotiation, approval, purchase-order generation, and delivery follow-up. These activities are often handled through disconnected tools and manual coordination, making decision evidence and workflow continuity difficult to preserve. Procura AI is a multi-agent procurement prototype that integrates these stages into a stateful workflow. It uses Python, FastAPI, LangGraph, SQLite, SQLAlchemy, WebSockets, structured supplier data, and a React/Vite interface. Supplier evaluation uses deterministic multi-criteria logic based on price, reliability, verification, and lead time. External offers require explicit pricing evidence and source provenance. A human approval step interrupts and resumes the workflow while preserving state. On a 20-case synthetic benchmark, Procura AI achieved 26.32% selection accuracy (5/19 eligible cases), compared with 10.53% (2/19) for a price-only baseline; both had 100% budget compliance, and Procura AI had 100% workflow success on applicable cases. These preliminary results do not establish production performance. RFQ, ERP, and logistics integrations use demo or local adapters; the system is a prototype, not a fully integrated production platform.

Index Terms— Multi-agent AI, procurement automation, supplier evaluation, LangGraph, human-in-the-loop, evidence-aware sourcing, workflow orchestration.

I. INTRODUCTION

Procurement is a sequence of interconnected activities involving requirement identification, supplier sourcing, quotation evaluation, negotiation, approval, purchase-order generation, and delivery tracking. When activities are managed independently, information can become fragmented, decision evidence can be lost, and recommendations can be difficult to verify. The Procura AI project documentation presents this coordination challenge as the motivation for an integrated workflow [6].

Large language models can assist with language-heavy tasks such as interpreting requests, summarizing supplier information, comparing offers, and drafting negotiation messages. Procurement decisions also involve budget constraints, supplier eligibility, approval authority, and traceable evidence. Procura AI therefore separates language assistance from deterministic application rules and human authorization.

The contribution is an end-to-end prototype combining specialized workflow stages, shared execution state, multi-criteria supplier evaluation, evidence-aware sourcing, and a human approval gate. It is a systems-integration prototype; it does not claim a new foundational model or fully autonomous purchasing.

II. PROBLEM STATEMENT AND OBJECTIVES

Procura AI aims to receive and validate requests; find supplier offers; rank eligible offers using multiple criteria; compare quotations; prepare negotiation or RFQ drafts; pause for human approval; create a purchase order after approval; initialize and update delivery tracking; and preserve context across workflow stages [6], [7]. The design emphasizes evidence, deterministic constraints, and human control. An LLM response alone cannot authorize spending.

The logical sequence is Request → Sourcing → Evaluation → Quotation Analysis → Negotiation → Approval → Purchase Order → Delivery Tracking. This conceptual flow is distinct from implementation choices such as LangGraph, FastAPI, SQLite, SQLAlchemy, and WebSockets.

III. LITERATURE REVIEW

A. Multi-Agent Vendor Evaluation

Joshi et al. propose a multi-agent LLM framework for vendor evaluation and risk-aware procurement decisions, considering dimensions such as financial health, risk, industry standards, and market conditions [1]. Their work motivates specialized analysis rather than one undifferentiated model. Procura AI applies this pattern across intake, sourcing, evaluation, quotation analysis, negotiation drafting, approval, PO creation, and delivery tracking.

B. LLMs for Procurement Process Automation

Brandtner and Hofer examine LLM use in procurement tasks including formal offer checking, information summarization, and bid comparison. Their reported experiment compares GPT-4o with human participants and discusses processing time while noting that strong human participants can outperform the model; bias, explainability, and operational risk remain concerns [2]. This supports using LLMs for text-intensive tasks while keeping application rules and approvals explicit.

C. Reliability of LLM Supplier Scoring

McKinley, Dohmen, and Castillo study agreement between generative AI and human procurement professionals on supplier bids. The cited study reports stronger alignment on compliance-related signals and more scoring volatility on competitive signals such as value-added propositions [3]. Its two-stage framing—AI-supported compliance screening with human judgment retained for competitive assessment—is consistent with Procura AI's human approval design.

D. Sustainable Supplier Selection

Nafei et al. present a framework combining LLMs, neutrosophic sets, and neural networks to process uncertain or conflicting natural-language judgments and aggregate structured evaluations for supplier ranking [4]. Procura AI's current ranking is deterministic; it does not implement the neural or neutrosophic method.

E. Generative AI in Buyer–Supplier Negotiation

Herold et al. investigate generative AI in buyer–supplier negotiations through experiments in which a ChatGPT-based chatbot acts as the buyer and human participants act as suppliers [5]. They report that a competitive strategy improves price discounts, payment terms, and negotiation speed, whereas a collaborative strategy improves supplier trust, satisfaction, and willingness to engage again. In Procura AI, the negotiation agent prepares drafts; the demo adapter does not send them to suppliers.

F. Research Gap

Prior studies examine individual tasks such as supplier evaluation, quotation analysis, negotiation, and process automation. Procura AI integrates these activities with PO generation, delivery tracking, and human authorization at the prototype level. The current evaluation does not establish superiority over the cited research systems. Table I summarizes the scope of the reviewed studies.

TABLE I. SCOPE OF REVIEWED STUDIES RELATED TO PROCURA AI

Ref.	Primary focus and approach	Relation to Procura AI
[1]	Multi-agent LLM framework for vendor evaluation and risk-aware decisions.	Supports specialized agents; Procura AI extends the pattern across a workflow.
[2]	GPT-4o versus human participants on offer checking, summarization, and bid comparison.	Supports text assistance with explicit controls.
[3]	LLM scoring volatility and comparison with human supplier evaluations.	Supports retaining human judgment for competitive assessment.
[4]	LLM, neutrosophic sets, and neural aggregation for sustainable supplier selection.	Context only; Procura AI uses deterministic ranking.
[5]	Chatbot buyer versus human suppliers; competitive versus collaborative strategy.	Informs drafting; no live supplier dispatch in the prototype.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the logical procurement sequence. At runtime, the React/Vite interface communicates with a FastAPI backend, which validates requests, accesses records through SQLAlchemy, and invokes the LangGraph workflow. LangGraph manages progression and checkpointed state; WebSockets support workflow updates. SQLite is the documented local persistence choice [6].

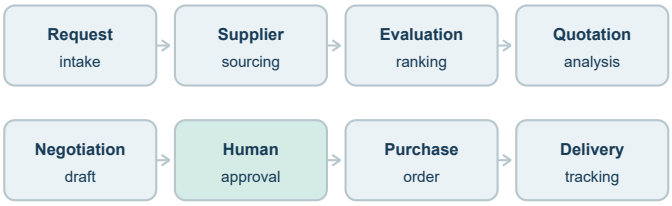


Fig. 1. Logical procurement architecture and local implementation boundary [6].

The public project documentation describes internal-first sourcing, with optional external supplier discovery when eligible internal offers are unavailable. An external offer is created only when an explicit price in the requested currency is extracted; source URL and price evidence are retained [6]. This does not guarantee that a third-party page is current or that a supplier is legitimate. Email/RFQ, ERP, and logistics actions remain demo/local adapters. Conceptual integration targets must not be mistaken for deployed services.

Rejection ends the purchase path without creating a PO. Approval enables PO generation and creation of a delivery record. Fig. 2 summarizes the pause-and-resume control flow.

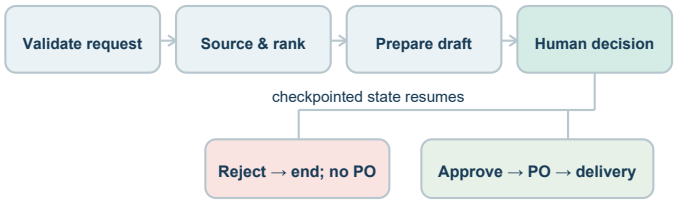


Fig. 2. Approval gate, rejection path, and resumable workflow state [6].

V. AGENT RESPONSIBILITIES AND WORKFLOW

1) Intake: receives a request, validates required fields and numeric constraints, and establishes shared procurement context. Language-model assistance may help interpret request text [6].

2) Supplier sourcing: checks structured internal offers first and may attempt external discovery when eligible internal offers are unavailable. Candidate offers are checked for explicit price, currency, budget, source URL, and supporting evidence. Missing pricing information or source evidence prevents an external quotation from being treated as valid [6].

3) Supplier evaluation: computes a deterministic score using price, reliability, verification, and lead time. Offers are ordered by score; lower price breaks score ties [6].

4) Quotation analysis: compares available offers and provides a top-ranked candidate for review. A recommendation is not purchase authorization.

5) Negotiation: prepares negotiation text or RFQ drafts. Drafts are not transmitted externally in the current demo [6].

6) Human approval: pauses for approval or rejection and resumes from saved state. The prototype's requester/approver name comparison is not secure identity verification or enterprise authorization [6].

7) PO and delivery: following approval, creates a local PO record and delivery record. The current ERP and logistics adapters do not submit to a live ERP or carrier [6].

VI. SUPPLIER EVALUATION: MATHEMATICAL FORMULATION

The project documentation describes supplier evaluation as a weighted multi-criteria score [6]:

$$S_i = w_P P_i + w_R R_i + w_V V_i + w_L L_i \quad (1)$$

Here, S_i is the score for offer i; P_i is its price-related score; R_i is supplier reliability; V_i is supplier verification; L_i is lead-time performance; and the w terms represent the relative importance of each criterion. The project description does not specify numeric weights for this general formulation; none are supplied here. Eligible offers are ranked in descending score, with lower price first when scores tie.

The application uses deterministic logic for supplier ranking, approval validation, PO creation, and workflow transitions. LLMs support language-intensive activities such as request understanding and negotiation drafting. The system is decision support, not a trained predictive model [6].

Algorithm 1. End-to-end procurement workflow

Input: procurement request	Output: updated workflow state
1. Receive request; validate fields; establish shared context. 2. Source eligible offers (internal first; optional external search requires price and source evidence). 3. Evaluate and rank offers (Algorithm 2); compare quotations. 4. Prepare negotiation/RFQ draft; pause for human decision. 5. If rejected, record rejection and end without a PO. 6. If approved, generate PO and create delivery record; continue tracking. 7. Return updated workflow state.	Workflow pauses at human approval and resumes from saved state. Rejection does not create a purchase order [6].

Algorithm 2. Supplier evaluation and ranking

Input: eligible supplier offers	Output: ranked offers
1. For each eligible offer i, evaluate price, reliability, verification, and lead time. 2. Calculate score S_i using (1). 3. Rank offers by score; when scores tie, rank the lower-priced offer first. 4. Return ranked offers.	Only eligible offers satisfying procurement constraints are considered [6].

VII. IMPLEMENTATION DETAILS

The documented stack includes Python, FastAPI, LangGraph, SQLite, SQLAlchemy-backed records, WebSockets, structured supplier data, and a React/Vite frontend [6]. LangGraph manages workflow state and approval endpoints. The LLM gateway can use an optional OpenRouter path; the local deterministic demo adapter remains available when no provider key is configured. No claim is made that a provider is required or that RFQ drafts are sent.

The project supports local workflow records for cases, supplier offers, comparisons, approvals, purchase orders, deliveries, and workflow events. The PO endpoint generates a local PDF. The current prototype is not a live enterprise integration and is not described as production-ready [6].

VIII. EXISTING APPROACH VERSUS PROCURA AI

Table II reproduces the conceptual contrast described by the project documentation [6]. It is not a controlled comparison of every feature in an enterprise procurement environment.

TABLE II. CONCEPTUAL APPROACH COMPARISON

Separated activities (conceptual)	Procura AI prototype
Individual procurement activities with manual coordination.	End-to-end stateful workflow.
Supplier evaluation may be price-focused.	Deterministic multi-criteria supplier evaluation.
Supplier data may lack traceable support.	Evidence-aware external sourcing requirements.
Approval handled separately.	Human approval pause and resume in workflow.
PO and delivery activities disconnected.	PO and delivery records in the same workflow.

IX. EVALUATION METHODOLOGY

The evaluation report compares the implemented Procura AI supplier-ranking approach with a separate price-only baseline on the same 20-case synthetic benchmark [7]. The dataset contains 60 synthetic offer rows; it is not actual user data, supplier quotations, or market research. The independent reference selection uses a predeclared rubric prioritizing acceptable supplier identity, verification, reliability, lead time, and then price [7].

The baseline sorts the same source-filtered offers by total price and deterministic ties. Procura AI uses the existing source helper and deterministic multi-criteria scorer. The evaluation used an isolated in-memory SQLite database and a deterministic demo LLM adapter. No OpenRouter or live Tavily network calls were made [7].

Selection accuracy is correct selections divided by cases with at least one eligible offer. C12 had no eligible offer, leaving 19 accuracy-eligible cases out of 20; therefore, 5/19 = 26.32% for Procura AI and 2/19 = 10.53% for the baseline. No-offer cases are excluded from this denominator [7].

Ranking is mean per-case Spearman correlation across 18 cases with at least two eligible offers. **Budget compliance** is the share of selected offers within budget. **Evidence coverage** counts raw synthetic offer rows with positive prices and nonempty source and price evidence. **Workflow success** counts eligible cases that paused for approval, resumed, and reached the expected post-decision state. **Processing time** is measured per-case selector time, excluding database seeding and workflow execution [7].

X. STRESS AND FAILURE-SCENARIO ASSESSMENT

The evaluation report contains scenario-level results but no defined load levels, concurrency counts, or quantitative load-test outputs. Table III distinguishes documented benchmark outcomes from scenarios that were not exercised; it does not claim new tests [7].

TABLE III. SOURCE-DOCUMENTED SCENARIOS AND EVIDENCE

Scenario	Source-supported outcome
Workflow success	19/19 eligible workflow cases paused, resumed, and reached the expected post-decision state.
Approval rejection	C13 reached REJECTED and created no PO; reported as PASS.
No valid offer	C12 ended as NO_MATCHING_OFFERS with no quote, approval, or interrupt; PASS.
Invalid supplier information	The internal source path retained a blank-name/out-of-range record; evaluation reports this validation scenario as FAIL.
External sourcing failure / missing price evidence	Live external sourcing was not exercised. External offers require explicit pricing evidence and source context; separate failure handling is not documented [6], [7].

XI. EXPERIMENTAL RESULTS AND DISCUSSION

Table IV and Figs. 3–6 summarize the supplied 20-case synthetic benchmark [7].

A. Selection and Ranking

Procura AI achieved higher selection accuracy than the price-only baseline (26.32% versus 10.53%) over the 19 cases with eligible offers. It matched the independent reference in C04, C07, C13, C16, and C19; the baseline matched in C07 and C08. C07 was shared. C12 had no eligible offer and is excluded from accuracy. Mean Spearman correlations were −0.1111 for Procura AI and −0.6944 for the baseline, across 18 multi-offer cases [7].

B. Constraints, Evidence, and Workflow

Both approaches reported 100% budget compliance and 93.33% evidence coverage. Evidence coverage is a shared-input measure (56/60 synthetic offer rows), not a measured production evidence-retention rate. Procura AI reported 100% workflow success (19/19 applicable cases); the baseline has no workflow equivalent [7].

C. Processing Time

Average selector time was 0.012375 s for Procura AI and 0.009147 s for the baseline, approximately 35% higher for Procura AI. These are per-case local benchmark averages, not load-test latency percentiles [7].

TABLE IV. REPORTED BENCHMARK RESULTS (20 SYNTHETIC CASES)

Metric	Price-only	Procura AI
Selection accuracy*	10.53%	26.32%
Mean Spearman ρ	−0.6944	−0.1111
Budget compliance	100%	100%
Evidence coverage	93.33%	93.33%
Workflow success	N/A	100%
Average selector time	0.009147 s	0.012375 s

*Selection accuracy is based on 19 cases with at least one eligible offer; the 20th case had no valid offer. Workflow success is N/A for the selector-only baseline.

Fig. 3. Selection accuracy (%)

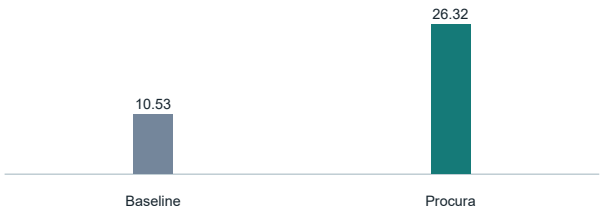


Fig. 3. Selection accuracy on eligible synthetic cases [7].

Fig. 4. Case-level correct selections



Fig. 4. Correct selections against the independent reference [7].

Fig. 5. Mean Spearman rank correlation



Fig. 5. Mean correlations; plotted bar lengths show absolute magnitudes, with signed values labeled [7].

Fig. 6. Average selector processing time (s)



Fig. 6. Average per-case selector time; lower is faster [7].

The small, synthetic benchmark does not establish statistical significance, real-world savings, generalization to live supplier markets, or production performance [7].

XII. LIMITATIONS, RISKS, AND FUTURE WORK

Procura AI is a local prototype. RFQ drafts are not sent to suppliers, ERP submissions are recorded locally, and requester–approver checks compare names rather than securely verifying identities [6]. The evaluation uses 20 synthetic cases and a policy-defined reference rubric; the rubric is not adjudicated procurement ground truth [7].

Supplier pages can be stale, incomplete, misleading, or adversarial. Source URLs and price evidence improve traceability but do not independently verify supplier identity or guarantee a valid commercial quotation. Production deployment would require supplier identity validation, quote freshness checks, total-cost normalization for taxes, shipping and currency, safe handling of retrieved content, and review paths for uncertain offers.

Future work includes enterprise, supplier-platform, and financial-system integrations; larger and more diverse benchmarks; monitoring of sourcing failures; and deployment improvements. Further evaluation should examine approval/rejection handling and checkpoint recovery. Production use would also require authenticated users, role-based access control, tenant isolation, secret management, backups, idempotent external actions, secure audit trails, and operational monitoring. These are recommendations, not currently implemented capabilities.

XIII. CONCLUSION

Procura AI integrates procurement intake, supplier sourcing, multi-criteria evaluation, quotation analysis, negotiation drafting, human approval, PO generation, and delivery tracking into a stateful workflow. The design combines evidence-aware sourcing and deterministic supplier evaluation with specialized stages, while reserving authorization for a human. The workflow can pause and resume with context preserved [6].

On the supplied synthetic benchmark, Procura AI matched the independent reference in 5 of 19 accuracy-eligible cases, versus 2 of 19 for a price-only baseline. Both achieved full budget compliance, and Procura AI completed all 19 applicable workflow cases. The small synthetic dataset and local demo integrations limit these findings. Procura AI should be understood as a working prototype for further validation and integration, not as a production-ready autonomous procurement system [7].

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